Core concurrent circular buffer

Created-at: 2026-09-13 03:42 +02:00. ETA: 25 minutes for integration, targeted correctness checks and publication; no performance run.

Scope and authorization. The user explicitly requests a qualified multithreaded circular buffer in Core, reusing Rust facilities where available. Expose Crossbeam's existing bounded MPMC implementation through Core util; do not maintain a second atomic algorithm. This request authorizes the scoped protected source and dependency edits. No existing callers require migration.

Files. hymeko_core/Cargo.toml, Cargo.lock, hymeko_core/src/util/mod.rs, hymeko_core/tests/circular_buffer.rs, SYMBOLS.md if generated output changes, this four-format plan, and the ignored reports/2026-09-13-concurrent-circular-buffer.md. Protected items: hymeko_core source and manifest.

Contract. Re-export ArrayQueue as CircularBuffer, with exactly pinned crossbeam-queue 0.3.14. Positive fixed capacity; push returns the rejected value when full; force_push explicitly evicts the oldest value; pop returns None when empty. Share across threads for Send payloads. Observations of length do not reserve slots. No fairness, real-time deadline, crash recovery or formal universal correctness claim. Allocation can fail; zero/excessive capacity can panic.

Qualification. Dedicated integration target only: FIFO and repeated slot reuse against VecDeque at capacities 1, 2, 3 and 7; overflow and eviction; exact ownership/drop accounting; concurrent delivery and producer order; small recorded operation histories checked against a sequential FIFO model with an iterative linearizability search and positive/negative oracle controls. No profiling, throughput loops, latency assertions or workspace aggregate. Run scoped Clippy, formatting, symbols, preflight and duplication checks.

Resources and risks. No performance budget is measured or claimed. Correctness fixtures are small, bounded workloads. Production risks include contention, starvation and unhandled rejection; callers must explicitly choose retry/backpressure or eviction. Tests do not exhaust hardware memory models, all schedules, stamp rollover or every target architecture.

Rollback. Revert this task's commit; no serialized format or existing behavior changes. Preserve pre-existing untracked LaTeX helper files.

Measurement addendum

Created-at: 2026-09-13 12:36 +02:00. ETA: about 60 minutes in total. Basis: 20 minutes bench source and accounting preflight, 3 minutes release build, one calibration cell (tens of seconds), a bounded matrix with at most 25 minutes of measured wall (48 Criterion throughput cells of roughly 3.5 s each and 48 latency cells of 1–5 s each), and 15 minutes for analysis, report and archival. If the calibration cell projects more than 25 minutes, the matrix is reduced before launch and the reduction is recorded.

Authorization. The user's current-task delegation to run targeted benchmarks of the published Core concurrent circular buffer authorizes these measurements only, plus the minimal benchmark source and manifest registration. For this addendum it supersedes the "no performance run" statement above. No production source, algorithm or dependency change; no profiler; no other benchmark.

Files. New `hymeko_core/benches/circular_buffer_bench.rs`; `hymeko_core/Cargo.toml` gains one `[[bench]]` entry with `harness = false` and `test = false` (protected, authorized above); this plan's four files; ignored `reports/2026-09-13-concurrent-circular-buffer.md` and `reports/2026-09-13-concurrent-circular-buffer/` (sidecar JSON, copied Criterion samples, `summarize.py`, CSV, plots, logs). `Cargo.lock` must stay byte-identical.

Default-off execution. `test = false` keeps ordinary `cargo test` from building the target. Additionally `main` returns before any measurement unless Cargo passed `--bench`, which also covers `--benches`/`--all-targets` test-mode invocations. The only launch is `cargo bench --manifest-path <core>/Cargo.toml -p hymeko_core --bench circular_buffer_bench --offline`, from the user's Temp directory with `CARGO_TARGET_DIR=<core>/target`.

Workload. Payload `u64` = producer index shifted by 40 bits or sequence. Implementations: Core `CircularBuffer` (Crossbeam `ArrayQueue`) and a bounded `Mutex<VecDeque>` with the same rejecting push, evicting force_push and FIFO pop contract; a sequential parity preflight compares both against each other before any timing. Modes: *rejecting* (push retried with a fixed spin-then-yield backoff until accepted) and *overwrite* (force_push; every evicted value returned and counted). Capacities 1, 64, 1024. Producers = consumers in {1, 2, 4, 8}; 8P/8C only when available parallelism is at least 17.

Round. One persistent worker pool per cell; threads are spawned once and wait on a start barrier between rounds, outside timing. Timed window: start-barrier release to the last consumer's completion. Producers split N items; a consumer stops only on an empty pop that began after it observed the producers-done flag.

Accounting (every round, warm-up included). Inserted = N. Rejecting: removed = N, evicted = 0. Overwrite: removed + evicted = N, rejected = 0. Wrapping sum and SplitMix64 checksum of inserted values equal those of removed and evicted values; buffer empty after the round. Rejected pushes and empty pops are counted separately and never enter a throughput numerator.

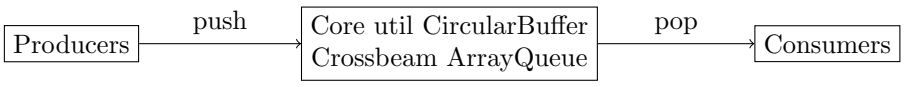
Throughput. Criterion `iter_custom`; one iteration is one produced item. Flat sampling, 1 s warm-up, 10 samples over 2 s (about 200 ms each). Per sample: transferred items/s (rejecting) or offered and delivered items/s (overwrite), reported as median, IQR and worst. Criterion `sample.json` iterations and times must equal the sidecar rounds exactly.

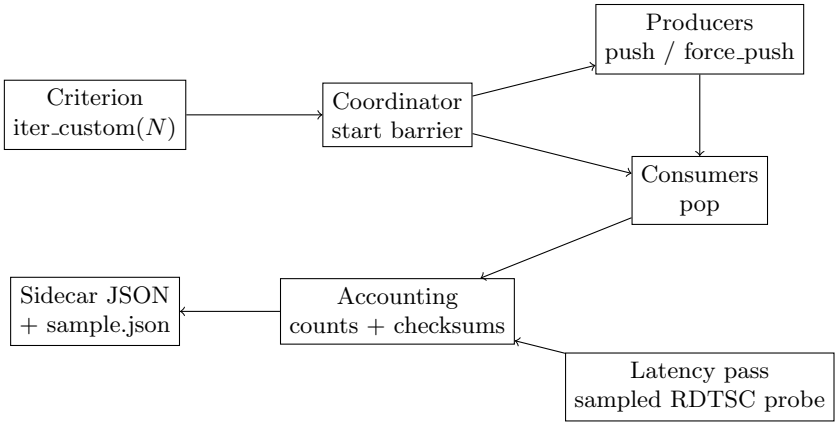
Latency. Separate pass with the same pool and a sampled probe. Every 8th buffer call per thread is timed with RDTSC calibrated against `Instant` (the Windows QPC resolution is 100 ns), classified by operation and outcome (accepted/rejected push, inserting/evicting force_push, removed/empty pop) into a log-linear histogram (16 sub-buckets per octave, at most 6.25% bucket width) with exact maximum. One warm-up and 5 measured repetitions of 2^{18} items: p50, p90, p99, p99.9 and maximum, median/IQR/worst over repetitions; instrumentation floor reported.

Budgets. Peak RSS below 1 GiB (fixed buffers, about 8 KiB per operation class per thread). Measured wall at most 25 minutes. No overnight run.

Risks. Hybrid P/E cores and Windows scheduling without affinity; Balanced power plan; background processes are recorded, not killed; the backoff policy determines failed-attempt counts; the latency probe perturbs contention; worker wake-up latency lies inside the timed window (tens of microseconds against 200 ms samples). Results do not generalize to other hosts, payloads or schedulers.

Rollback. Remove the benchmark file and its `[[bench]]` entry; no library behavior changes.




Criterion
iter_custom(N)
Coordinator
start barrier
Producers
push / force_push
Consumers
pop
Accounting
counts + checksums
Sidecar JSON
+ sample.json
Latency pass
sampled RDTSC probe